

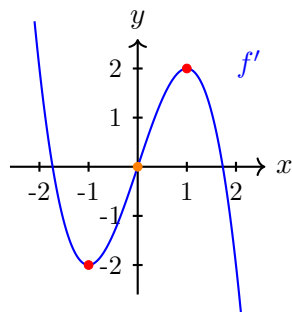
Name: _____

Mark: _____ / 7

Mini-math Div 3/4: Wednesday, September 16, 2025 (5.1-5.7) - 14 minutes

1. (1 point) A differentiable function f is defined on the closed interval $[-2, 4]$. Which of the following statements must be true?
- A. If f has a local maximum at $x = 4$, then $f'(4) = 0$
 - B. If $f'(c) = 0$ for some $c \in (-2, 4)$, then f has a local extremum at $x = c$
 - C. If f has a local minimum at $x = 2$, then $f''(2) > 0$
 - D. If f has a local maximum at $x = 1$, then $f'(1) = 0$

2. (1 point) The graph of f' , the derivative of a twice-differentiable function f , is shown below. How many inflection points does the graph of f have on the interval shown?



- A. 0
- B. 1
- C. 2
- D. 3

3. (1 point) Let $f(x) = x^3 - 6x^2 + 9x$. On the interval $[0, 4]$, the Mean Value Theorem guarantees the existence of a point c in $(0, 4)$ for which $f'(c)$ equals the average rate of change of f on $[0, 4]$. What is the value of c ?

A. 1

C. $2 + \frac{2\sqrt{3}}{3}$

B. 2

D. $2 - \frac{2\sqrt{3}}{3}$

4. (1 point) The derivative of a function h is given by $h'(x) = (x + 2)(x - 1)^2(x - 4)$. At which of the following values of x does h have a local minimum?

A. $x = -2$

C. $x = 4$

B. $x = 1$

D. h has no local minimum

5. (1 point) Let $f(x) = x^4 - 4x^3$. Use the First Derivative Test to classify the critical points of f .

6. (2 points) Find the absolute maximum and absolute minimum values of $g(x) = x + 2 \sin x$ on the closed interval $[0, \pi]$. Show your work.